

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460092.7601 +/- 0.0046 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.08 +/- 0.025
Transit depth (Rp/Rs)^2	0.64 +/- 0.4 [%]
Orbital Inclination (inc)	85.7 +/- 2.72
Airmass coefficient 1 (a1)	1.184 +/- 0.015
Airmass coefficient 2 (a2)	-0.1175 +/- 0.0073
Scatter in the residuals of the lightcurve fit is	1.24 %
Best Comparison Star	None
Optimal Aperture	4.07
Optimal Annulus	7.8
Transit Duration (day)	0.0872 +/- 0.0074

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.08 ± 0.025 vs 0.126 (deviation 1.84σ)
Duration fit vs published	0.0872 d vs 0.0925 d (deviation 0.72σ)
Mid-time O–C	-2.6 min
Depth SNR	3.2
Residual scatter	1.24 %

Light curve

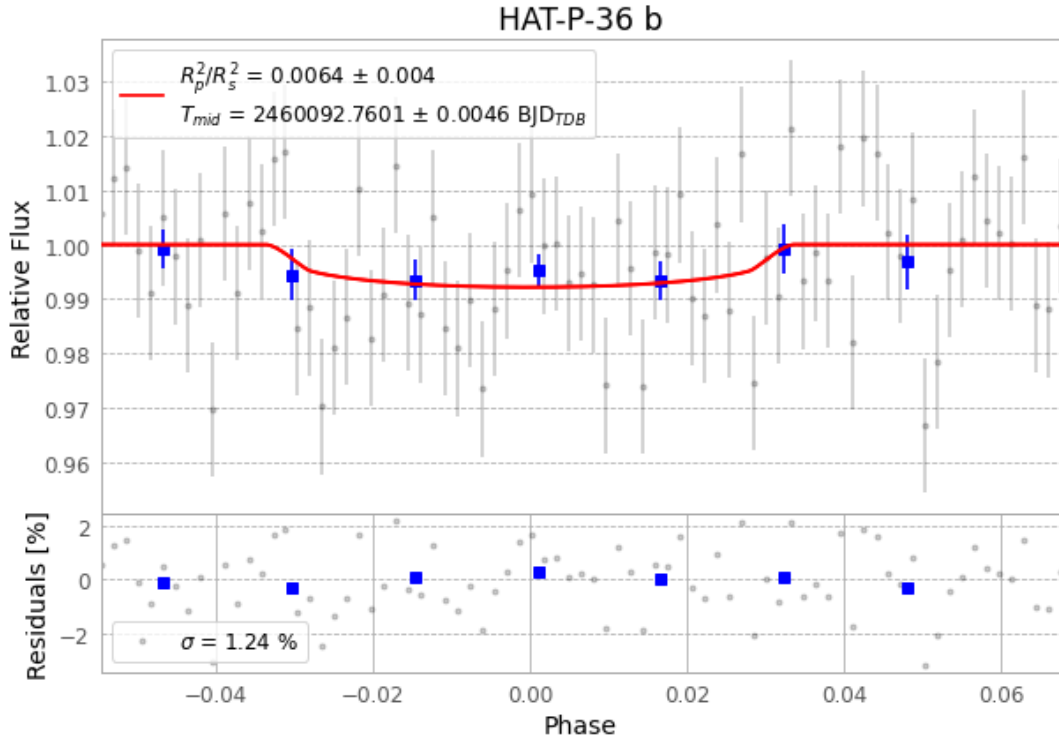


Figure 1 — FinalLightCurve_HAT-P-36 b_2023-05-27.png (SHA-256 cfd79acaea5cb214...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [294, 377], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4d350f80a99f29a2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

79 FITS frames (52 MB), HATP-36230528042715.FITS → HATP-36230528082115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-230528.log (16f9d5ff1112...), FinalLightCurve_HAT-P-36 b_2023-05-27.png (cfd79acaea5c...), FinalLightCurve_HAT-P-36 b_2023-05-27.csv (02fe8018cff6...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 05:44Z.